Dhirubhai Ambani University
Technology
Formerly DA-IICT

# CT204—Signals and Systems (3-0-2-4)

## AY: 2025-2026

**Course Placement:** Signals and Systems is a core UG course for 2nd year (3rd semester) B.Tech ICT and ICT+CS programs.

**Course Format:** CT204-- **3 hours' lecture** and **2 hours' lab** every week.

**Course Objective:**

The course serves as a foundation/introduction to the area of signal and image processing. The objective of the course is to introduce the concept of signal and system modeling and to introduce a few paradigms for modeling and analyzing signals and systems. The core of this course deals with Linear systems and Linear time-invariant systems. Several related transforms: DFT, DTFT, Fourier Series, CTFT will be taught as instances of the similarity transformation, while also introducing the Laplace and z-transform. LTI systems will be analyzed using these transforms.

**Course Content:**

1. Introduction to signals: Continuous time and discrete time signals; Periodic and aperiodic signals; Even and odd signals; Signal energy and power; Unit step, unit impulse, ramp, exponential, sinusoids; transformations of the independent variable
2. Introduction to systems: linearity, time invariance, causality, memory, stability, Linear and time invariant system, impulse response, convolution integral
3. Frequency: Definition of frequency, Fourier series, Continuous time Fourier transform, Discrete time Fourier transform, Convolution Theorem, Parseval's relation, Convergence
4. Sampling: Sampling theorem, Properties, Frequency sampling, Circular convolution, Discrete Fourier transform
5. Laplace transform: Generalization of CTFT to LT, Poles and zeros, region of convergence, Properties
6. Z-transform: Generalization of DTFT to ZT, Poles and zeros, region of convergence, Properties
7. Random Processes: Auto correlation function, power spectral density, Stationary processes.

**Readings:**

1. Lecture Notes/slides
2. A. V. Oppenheim, A. S. Wilsky and S. H. Nawab, “Signals and Systems,” 2nd edition, Prentice-Hall of India, 1999.
3. B. P. Lathi, “Linear Systems and Signals,” Berkeley-Cambridge Press, 1992.
4. Haykin, Simon, and Barry Van Veen, “Signals and Systems,” John Wiley & Sons, 2002.

**Assessment Method (Tentative):**

- 1st In-semester examination – 15%
- 2nd In-semester examination -- 25%
- End-semester examination – 40%
- Lab Experiments, Reports, and Lab Exam – 20%

**Course Outcomes:**

After completion of this course, students should be able to:

- Develop signals and systems concept for modeling physical processes for different applications **[P1, P12]**
- Analyze practical systems for different inputs **[P2]**
- Design necessary filters (both analog and digital) for practical applications **[P3]**
- Compute convolution between (simple)sequences/functions, frequency response of a simple LTI system, Z-transform/Laplace transform of simple sequences/functions, the pole-zero plot and understand its implications for the system. **[P1, P5]**
- Work in a group for a laboratory experiment, and present their results **[P10]**.

**Mapping of Course Outcomes to Program Outcomes:**

| P1 | P2 | P3 | P4 | P5 | P6 | P7 | P8 | P9 | P10 | P11 | P12 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| X | X | X |  | X |  |  |  | X | X |  | X |

**Lecture Schedule:**

| Sl. No. | Description | | No. of Lectures |
|---|---|---|---|
| **1** | **Signals** | | **6** |
| | 1.1 | Continuous time and discrete time signals; Periodic and aperiodic signals; Even and odd signals; Signal energy and power | 3 |
| | 1.2 | Basic signals (unit step, unit impulse, ramp, exponential, sinusoids), transformations of the independent variable (shifting, scaling, reversal) | 3 |
| **2** | **Systems** | | **6** |
| | 2.1 | Different types of systems | 1 |
| | **2.2** | Properties: linearity, time invariance, causality, memory, stability | 2 |
| | **2.3** | Linear and time invariant system, impulse response, convolution integral | 3 |
| **3** | **Fourier Series** | | **6** |
| | **3.1** | Representation of periodic signals | 3 |
| | **3.2** | Convergence | 1 |
| | **3.3** | Properties | 2 |
| **4** | **Continuous Time Fourier Transform** | | **5** |
| | **4.1** | Representation of aperiodic continuous time signals | 2 |
| | **4.2** | FT of Periodic Signals | 1 |
| | **4.3** | Properties | 2 |
| **5** | **Discrete Time Fourier Transform** | | **3** |
| | **5.1** | Representation of aperiodic discrete time signals | 1 |
| | **5.2** | Properties | 2 |
| **6** | **Sampling** | | **7** |
| | **6.1** | Sampling Theorem | 2 |
| | **6.2** | Reconstruction of a signal from its samples | 1 |
| | **6.3** | Effect of Under-sampling | 1 |
| | **6.4** | Frequency sampling, Circular convolution | 1 |

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|---|---|---|---|
| | **6.5** | Discrete Fourier transform | 2 |
| **7** | **Laplace Transform** | | **4** |
| | **7.1** | Generalization of CTFT to LT | 1 |
| | **7.2** | Poles and zeros, region of convergence | 1 |
| | **7.3** | Properties | 1 |
| | **7.4** | Butterworth filter | 1 |
| **8** | **Z Transform** | | **3** |
| | **8.1** | Generalization of DTFT to ZT | 1 |
| | **8.2** | Poles and zeros, region of convergence | 1 |
| | **8.3** | Properties | 1 |
| **9** | **Random Processes** | | **2** |
| | **9.1** | Autocorrelation function, Power Spectral Density | 1 |
| | **9.2** | Stationary processes | 1 |